

EduTrack AI: Capstone Project

Sustainable Development Goal: SDG 4 – Quality Education

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1. Selected SDG and Reason for Selection

Goal 4 – Quality Education: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.

Reason for Selection: Education is the foundation of sustainable development. However, many educational systems rely on "lagging indicators" (final grades) to identify struggling students. By the time a student fails a final exam, it is often too late for effective intervention. This issue is particularly severe in overcrowded classrooms or under-resourced schools where individual performance tracking is manual and slow. I selected SDG 4 because AI can bridge this gap by providing **Early Warning Systems** that identify at-risk students *during* the term, allowing for timely, data-driven support.

2. Problem Statement

- **What is the problem?:** Lack of automated, real-time tracking for student academic risk, leading to high failure rates that could have been prevented with early intervention.
- **Where does this problem exist?:** In schools and universities where large student-to-teacher ratios make it impossible for educators to manually analyze performance trends for every student across multiple subjects.
- **Who is affected?:** Primarily students at risk of academic failure, but also teachers who lack actionable insights and administrators who cannot effectively allocate support resources.
- **Why is it serious?:** Academic failure leads to increased dropout rates, loss of motivation, and long-term socio-economic disadvantages, hindering the progress of SDG 4.
- **What happens if it is not solved?:** Educational inequality will widen as students without private tutoring or individual attention continue to fall through the cracks of a reactive system.

3. Proposed Solution

EduTrack AI is a full-stack AI platform designed to predict academic risk and automate intervention planning. It uses a **Machine Learning soft-voting ensemble** (Random Forest + Logistic Regression) to analyze mid-term performance data and predict the probability of failure.

- **How it works:** The system ingests data like attendance rates, study hours, and mid-term marks. It processes these through a trained classifier that flags students as "High", "Medium", or "Low" risk.
- **Intervention:** It identifies the "Weakest Subject" for each student and uses rule-based logic to recommend a specific increase in weekly study hours (e.g., +5 hours/week) to mitigate the risk.
- **Impact:** By converting raw data into "Actionable Insights", it shifts the educational approach from *reactive* (fixing failures) to *proactive* (preventing them).
- **Users:** Teachers (for classroom monitoring), School Administrators (for resource allocation), and Students/Parents (for personalized study plans).

4. Project Features

Overview Dashboard

Real-time visualization of class-wide metrics (Avg Attendance, Pass Rates) using interactive Plotly charts and KPI cards.

Individual Student Lookup

A detailed profile search that breaks down a student's risk probability per subject and compares their scores to class averages.

Risk Prediction Sandbox

A simulator where teachers can input hypothetical parameters (e.g., "What if this student increases study hours to 10?") to see the predicted change in risk.

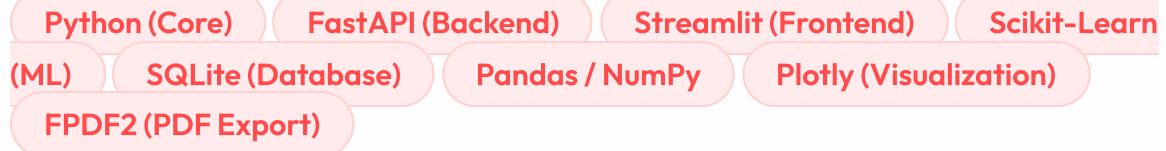
Automated Intervention Plans

Logic-driven recommendations that pinpoint the weakest subject and prescribe a personalized study hour adjustment.

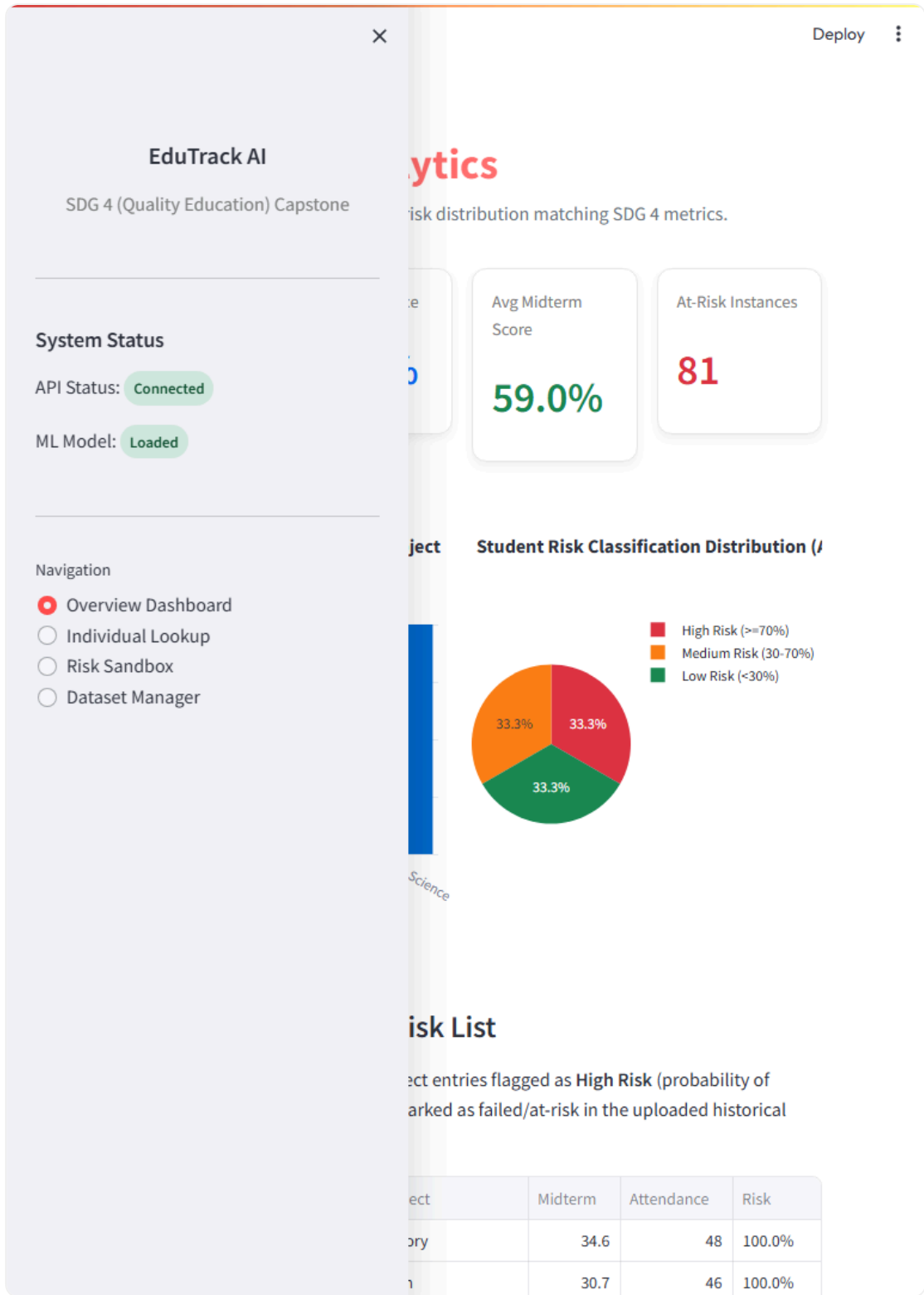
Report Generation

One-click export of student performance records and intervention plans into formatted PDF and CSV files for distribution.

5. Technology Stack



6. Project Screenshots (Proof of Work)



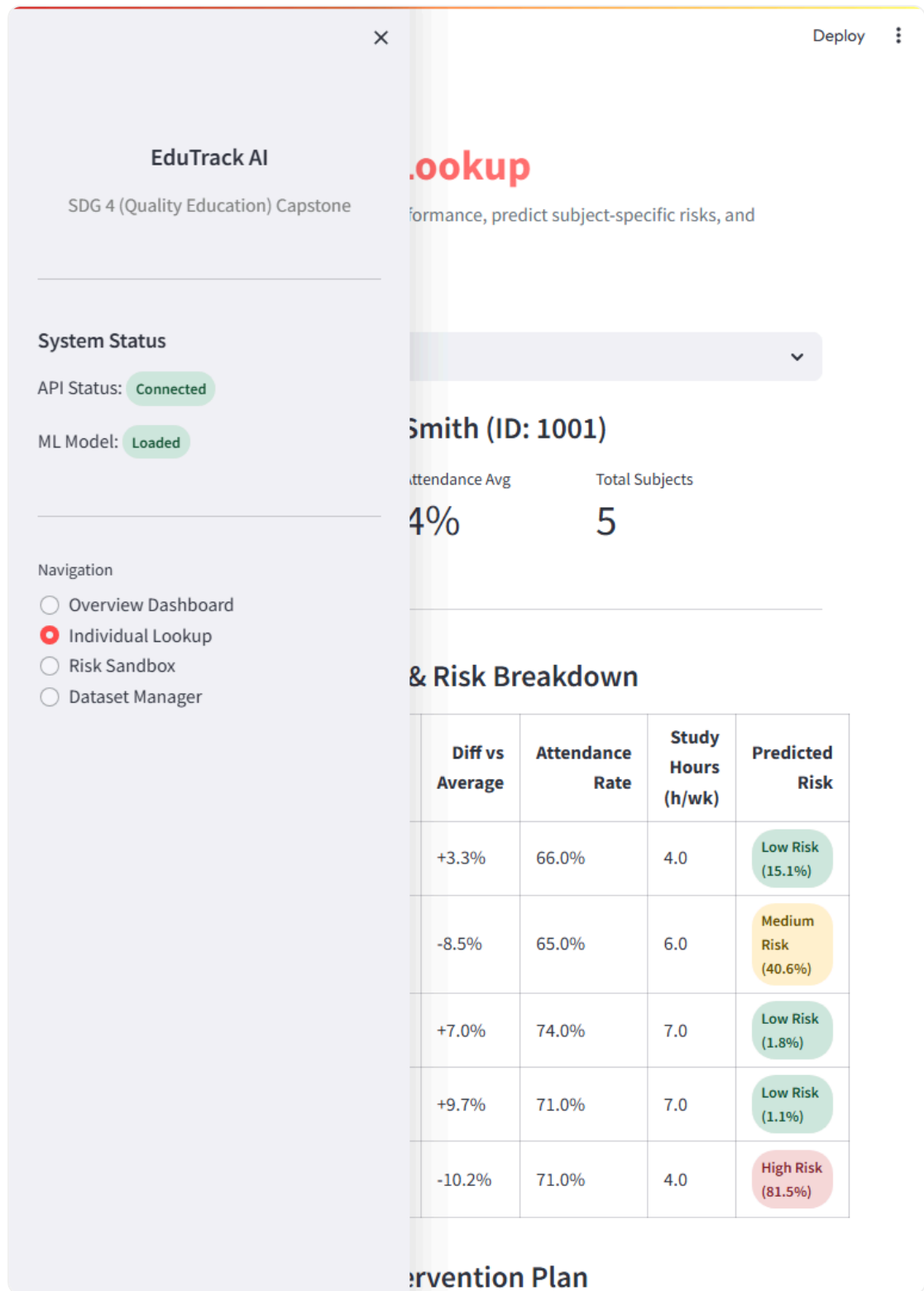
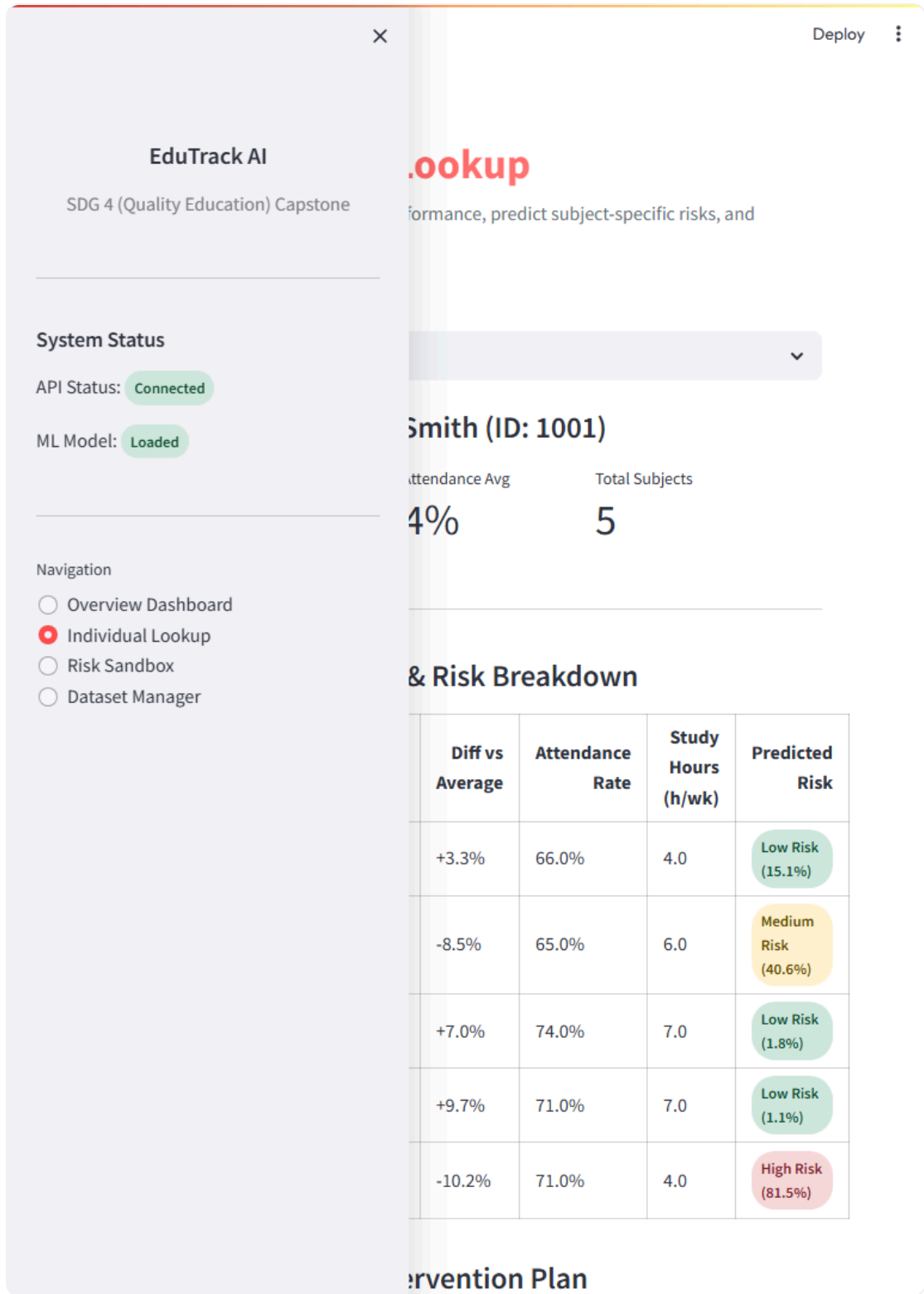


Figure 2: Individual Student Performance Lookup and risk probability breakdown.



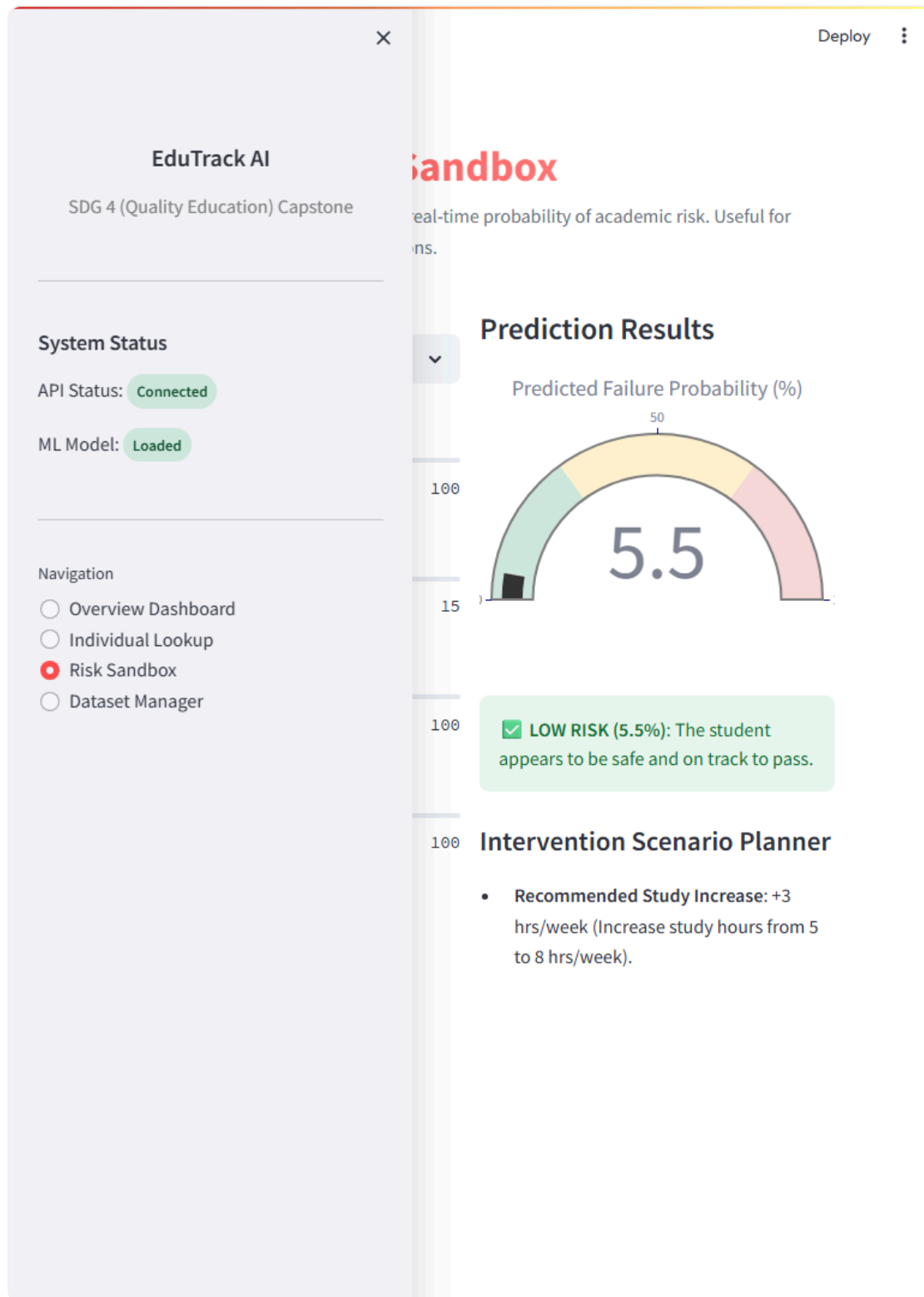


Figure 4: Risk Prediction Sandbox providing real-time probability gauging for hypothetical scenarios.

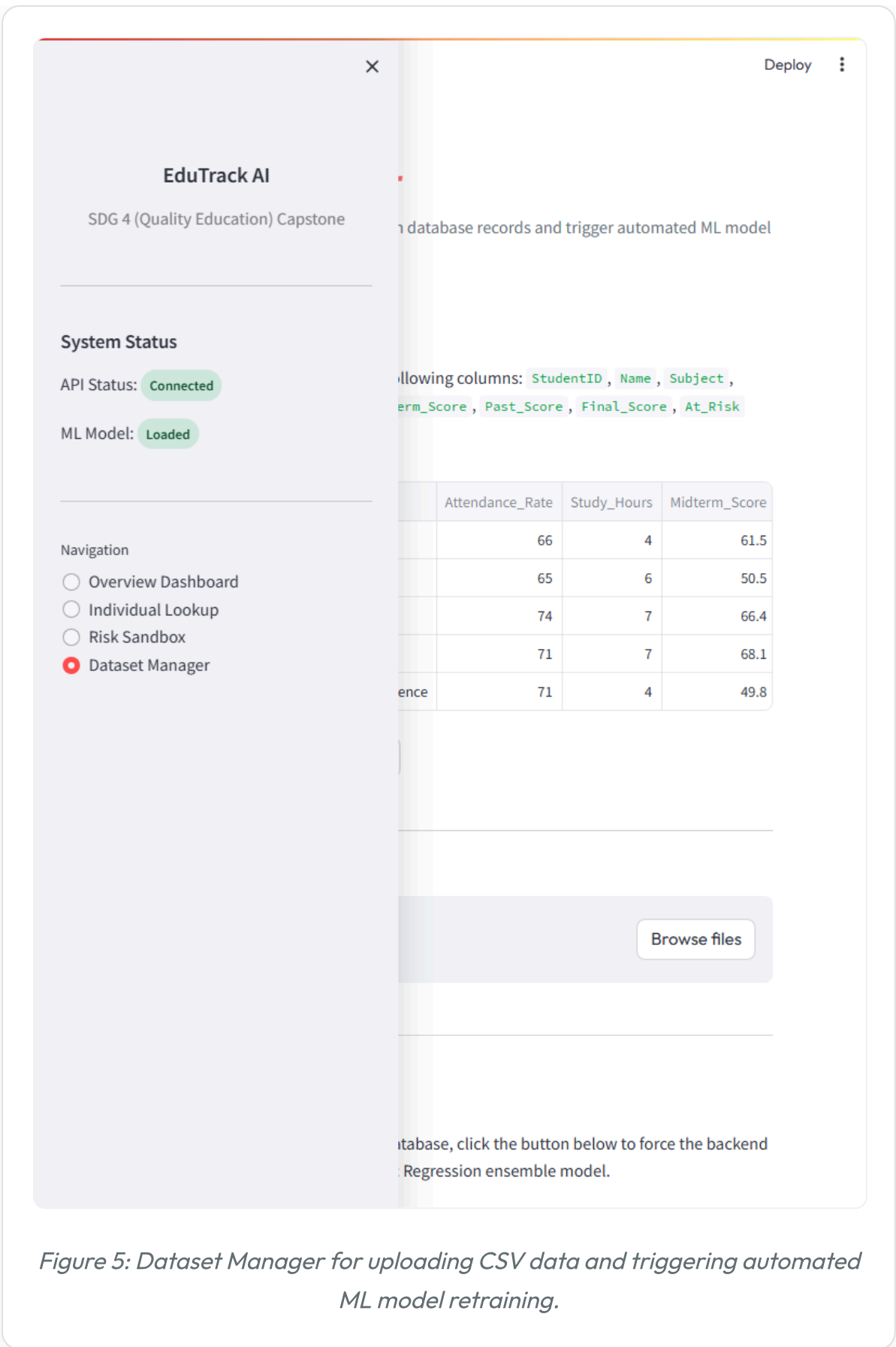


Figure 5: Dataset Manager for uploading CSV data and triggering automated ML model retraining.

7. Open Source Repository

The complete source code, including the ML training scripts, FastAPI backend, and Streamlit frontend, is available on GitHub:

<https://github.com/vijaybarhate/edutrack-ai>

8. Future Scope

- **Generative AI Integration:** Integrate a LLM-based tutor (e.g., Google Gemini) to provide immediate tutoring on the student's weakest subject directly within the app.
- **Mobile Companion App:** Develop a mobile version for parents to receive push notifications if their child's risk probability crosses a certain threshold.
- **LMS Connector:** Build API hooks for Google Classroom and Moodle to automatically sync grades and attendance data in real-time.
- **Advanced Modeling:** Incorporate socio-economic and behavioral factors into the ML model for a more holistic prediction of student success.

